



Assessment of Chinese paddy-soil structure using X-ray computed tomography

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ABSTRACT

Soil structure is known to affect water and solute movement and to characterize soil quality. For paddy fields, the soil structure has not been explored nor discussed in detail with respect to rice crop nutrient uptake and root development. The objectives were to visualize the size and connectivity of structural pores including arrangement of aggregates and to quantify the vertical bulk density distribution in the upper soil horizons affected by cultivation. The experimental paddy fields, located in subtropical SE-China in the province Jiangxi, were 20-years (Y) and more than 100-years old (O). Undisturbed soil columns of 10 cm diameter (5–42 cm and 0–38 cm depth) including the plough pan transition down to the subsoil were scanned using a medical X-ray computed tomograph (CT) with spatial resolution of 0.25 mm in horizontal and 1 mm in vertical direction. Vertical bulk density profiles were calculated from Hounsfield Units (HU) and gravimetric water contents measured in 10 mm and particle densities in 20 to 50 mm vertical intervals. Secondary pores were separated using 2 HU-threshold values and described by 3D plots of 'air-filled' and 'low-density' regions; the matrix structures were analyzed by 2D CT-images. Vertical root channels reached throughout the compacted horizons down to the bottom of samples from both fields. Aggregated matrix structures were found in the subsoil of both samples and horizontal cracks were identified between the cultivated horizon and the pan for Y. The compacted horizon was significantly deeper and larger for the sample from field O as compared to that from field Y, and the bulk density for sample O decreased abruptly in contrast to Y where density decreased gradually with depth. The contrasting bulk density distributions suggest that plough pan compaction continues and did not reach a 'steady-state', especially for sample Y. The results indicate the importance of the management system on bulk density and for soil structure formation.

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1. Introduction

In paddy rice fields, the soil structure of the topsoil is homogenized by puddling and a plough pan is formed by mechanical compaction and clogging of pore spaces by dispersed particles (Liu et al., 2005). The plough pan is the major factor controlling the water percolation by enabling ponding of water (e.g., Tuong et al., 1994; Chen et al., 2002; Liu et al., 2003). The hydraulic conductivity of the cultivated (puddled) horizon was found to be relatively small (e.g., Wopereis et al., 1994; Tourné et al., 2006). In structured soil, the secondary pores, like root channels or inter-aggregate spaces between aggregated matrix elements, control the hydraulic properties depending on their pore size distribution and pore connectivity (e.g., Mohanty, 1999; Mori et al., 1999; Köhne et al., 2002). Lin et al. (1999) and Pachepsky and Rawls (2003) showed that morphometric indices may rule the soil hydraulic properties rather than texture, bulk density or organic matter content. For use in modelling water and solute fluxes, soil texture, and bulk density was combined with arrangement and size of

secondary pores and soil aggregates to characterize soil horizons with different hydraulic properties (e.g., Tourné et al., 2006). By stereo-radiographic investigation of drainage out of 125 cm³ samples from the subsoil (i.e., at 50 cm depth), Mori et al. (1999) found that drainage from Japanese paddy soils was largely ruled by the soil structure. Ishiguro (1991) used a coaxial cylindrical model to simulate Br⁻ transport through hard pans with rice roots of paddy fields. The model was used to estimate pore densities of 1.6 to 3.1 cm⁻² with radii of 0.33 to 0.55 mm from breakthrough curves, without using morphological data.

Sander and Gerke (2007) demonstrated by using dye tracer, that the plough pan of an experimental paddy field in southeast China was strongly heterogeneous and allowed preferential flow (e.g., Beven and Germann, 1982) towards the shallow groundwater; the plough pan enhanced horizontally-redistributed flow prior to downward channelling along cracks and larger biopores.

Soil structure is further considered as an important parameter describing soil quality (e.g., BMEF, 1991), which affects not only trafficability, workability and water retention, but also influences soil/plant interaction (Munkholm, 2000) like rice root development and transport of water, air and chemicals (Anders et al., 2005). Knowledge about interaction between soil structure and plants and about

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structure effects on flow processes is still limited and should be given more attention (Gregory, 2006).

The structural characterization of paddy soils is an important topic because research is ongoing to increase the productivity by development of new technology and the use of machines, worldwide (Vethaiya et al., 2003). A change from traditional management of paddy rice fields with water buffalos to machines (Zhang et al., 2006) is likely to come along with changes of soil structure (Liu et al., 2005). The intensification of agricultural production for a growing population leads to increased application of fertilizers and pesticides, and non-point source pollution of groundwater and water bodies becomes an urgent problem (Zhu et al., 2000; Tong, 2003; Zhang et al., 2005). Lal (2000) stressed that management of soil structure and tilth, based on studies on dynamics of physical properties in puddled soils of rice-based cropping systems (i.e., through the development of indicators of soil physical quality among others), can be considered as the issues for the 21st century.

The X-ray computed tomography (CT) technique permits non-destructive investigations of the soil pore system and results in three-dimensional (3D) maps of X-ray attenuation (De Gryze et al., 2006). The attenuation expressed in values of the Hounsfield unit (HU) correlates with local composition of solid, water and air phases and allows the small-scale 2D and 3D determination of soil physical properties (Schrader et al., 2007). For estimating 3D soil bulk density distributions, the X-ray attenuation further requires information on water contents and solid particle densities at a representative local scale (Rogasik et al., 2003). The spatial resolution of medical CT scanners is currently in a range of about 500 μm , which is low

Table 1

Mean values of textural classes and standard deviations (in parenthesis) from three replicates) for samples from the younger (Y) and the older (O) paddy field and for particle-size classes 'sand' (2.0–0.063 mm), 'silt' (0.063–0.002 mm), and 'clay' (<0.002 mm)

Sample	Depth interval cm	Sand g kg ⁻¹	Silt	Clay
Y	0–5	–	–	–
	5–10	275 (4)	378 (2)	348 (4)
	10–15	276 (3)	379 (2)	344 (1)
	15–20	271 (3)	383 (4)	344 (2)
	20–27	222 (3)	394 (3)	384 (1)
	27–32	214 (2)	400 (0)	387 (2)
	32–37	215 (0)	400 (2)	386 (2)
	37–42	202 (4)	412 (1)	386 (5)
O	0–5	348 (2)	438 (3)	214 (2)
	5–10	362 (4)	414 (5)	224 (3)
	10–15	399 (1)	395 (3)	206 (3)
	15–20	414 (1)	403 (4)	184 (3)
	20–25	365 (2)	421 (3)	214 (1)
	25–30	315 (1)	445 (4)	240 (4)
	30–35	262 (3)	426 (3)	313 (1)
	35–38	270 (1)	409 (1)	321 (0)

compared to about 200 nm of nano-CT; however, medical CT facilitates scanning of larger (10^1 to 10^2 cm) samples, while scanning at higher spatial resolution is limited to smaller (10^{-1} to 10^0 cm) samples (Cnudde et al., 2006).

Our basic hypothesis is that characteristic soil structures, previously identified as cracks, root channels and worm burrows, are responsible for the occurrence of preferential flow in paddy soils. Structures may change with the duration of cultivation as a 'paddy' rice field due to repeated soil puddling and plough pan compaction. Except for the puddled topsoil, amount and relative distributions of each of the structures are assumed to decrease. The specific hypothesis here is that X-ray computed tomography can be used to characterize the relevant and visually observed structures that can then quantitatively be described and potentially used for numerical modelling.

The aim of this study was to visualize and describe the size and arrangement of structural pores, the arrangement of aggregates, and the bulk density distribution in the puddled topsoil, the compacted plough pan, and the subsoil of a 20-y and a 100-y old paddy soils by using X-ray computed tomography. Visually observed structures are to be discussed by quantitative parameters from X-ray scanning and data processing. Changes of soil structure with depth are discussed by comparing morphological observations with bulk density distributions and impacts on water flow that have been demonstrated by Sander and Gerke (2007) and Janssen and Lennartz (2007).

2. Material and methods

Soil sampling was carried out in the typical subtropics of the Peoples' Republic of China in a small watershed named Sunjia Catchment. The experimental site is located approximately 5 km northwest of the Ecological Experimental Research Station of Red Soil in Yingtan, Jiangxi Province, which belongs to the Chinese Academy of Sciences in Nanjing. Soils developed from quaternary clay. Two paddy fields of different age, in the following referred to as field O (for old) and field Y (for young), were chosen for soil structural investigations by CT-scanning.

Soil of field Y was classified as Stagnic Cambisol according to FAO classification scheme (FAO, 1998). The texture of the topsoil down to 42 cm is clay loam. The compacted plough pan started at about 14 cm depth and gradually changed to a cemented Bw2 horizon that reached down to about 30 cm. The compacted plough pan and the cemented Bw2 horizon below the plough pan showed prismatic and blocky



Fig. 1. Procedure of taking cylindrical samples (diameter 10 cm, heights 18 and 17 cm). Preparation of the sample using a knife to fit to Plexiglas cylinders and creation of cylindrical sample shapes by using a circle-wise guided hinged knife (close-up picture at the top and the entire device at the bottom).

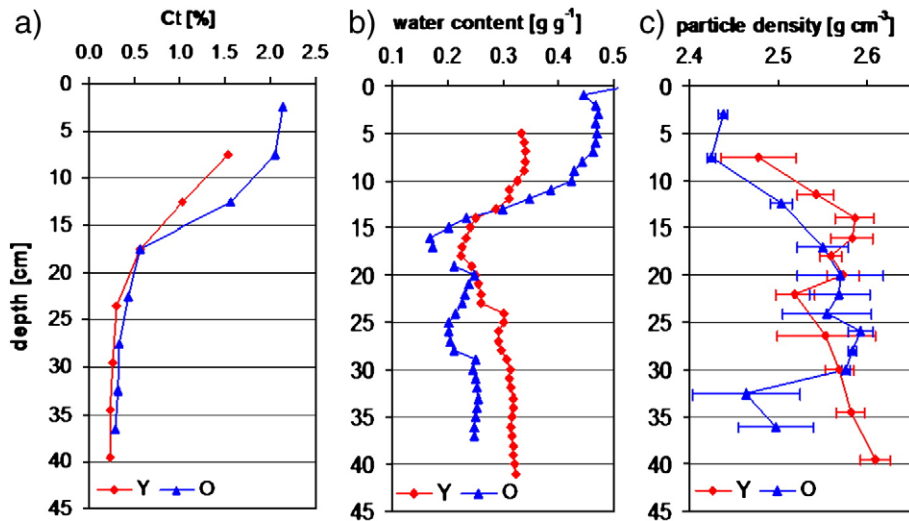


Fig. 2. Mean values of a) total carbon content, b) gravimetric water content, and c) particle density versus soil depth for samples Y (red) and O (blue). The error bars in c) indicate standard deviations.

macro-aggregates, which included strongly-embedded granular sub-units and crumbs (definitions according to Munkholm, 2000; Ball and Douglas, 2003). A platy horizontal structure was found on the top of the plough pan; whereas the subsoil appeared less compacted and had granular structure (with approximately spherical units of 2 to 5 mm). For a period of at least 20 years, the annual cropping system has been early rice (*Oryza sativa* L.) – early rice – fallow. Before that, pear trees were raised at this site. During cropping season 2004,

peanuts (*Arachis hypogaea* L.) were exceptionally cultivated for the first time. Here, puddling (i.e., mechanical mixing of water-saturated soil) was carried out twice a year by a rotary implement attached to a water buffalo (see Janßen et al. (2006) and Sander and Gerke (2007; here Y was named field 2) for further information).

For field O, soil was classified as Dystric Fluvisol (FAO, 1998). Textural class was loam down to 20 cm depth and clay loam from 20 to 45 cm soil depth. The soil beneath the cultivated horizon (at a depth of

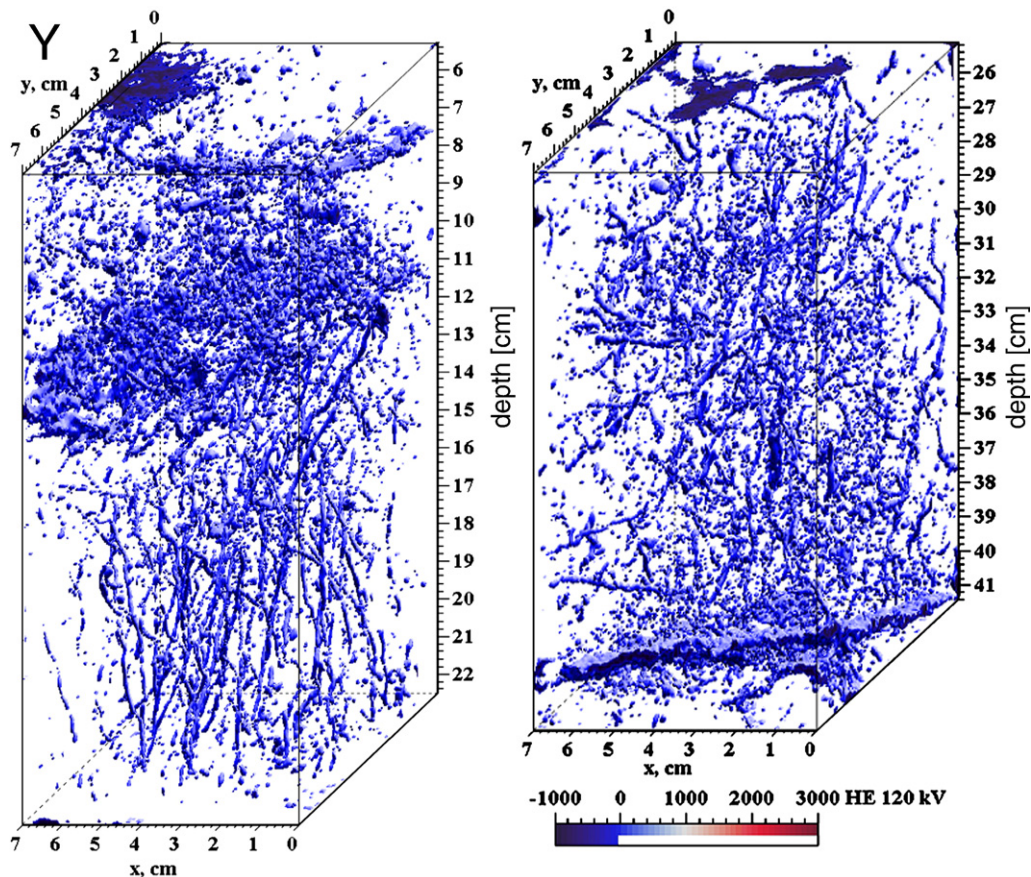


Fig. 3. The 3D distribution of 'low-density' pore regions (HU- (HE 120 kV) values < -43) in sample Y, for depths 5–22 cm (left) and 25–41 cm (right).

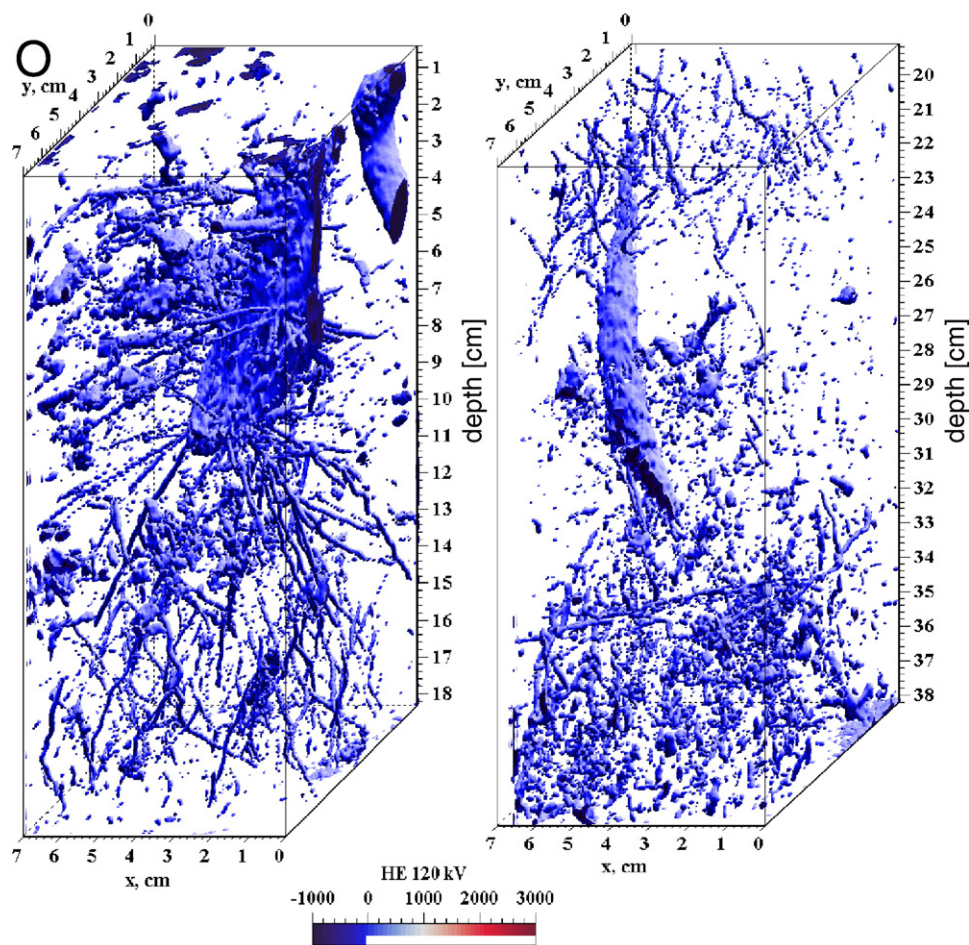


Fig. 4. The 3D distribution of 'low-density' pore regions (HU- (HE 120 kV) values < -43) in sample O, for depths 0–18 cm (left) and 19–38 cm (right).

about 13 cm) was more uniformly compacted down to 30 cm and showed a coarse platy structure. At 30 to 45 cm depth, macro-aggregates of larger horizontal than vertical extension were formed from clusters of round aggregates of about 1 to 5 mm. This paddy field (O) was cultivated with early rice – early rice for more than 100 years (Janßen et al., 2006; Janssen and Lennartz, 2007).

For investigations of the soil structure by X-ray computed tomography, large undisturbed soil monoliths (20 cm $\times$ 25 cm) at 5 to 23 cm and 25 to 42 cm depth for field Y and at 0 to 18 cm and 19 to 38 cm depth for field O were sampled on August 6 and 10 in 2004. Before sampling, a plot of about 3 m² had to be drained by construction of small soil bunds in field O. On this place, a free-standing soil column was created with a shovel. The samples were packed into cling film, put in Styrofoam-cushioned aluminium boxes, and were transported to Germany. From these soil blocks, undisturbed soil core samples (10 cm diameter, 17 and 18 cm heights) were created by using a circle-wise guided hinged knife (Fig. 1) and inserted into Plexiglas cylinders.

Soil material for determination of gravimetric water content and particle density was sampled centimeter-wise at several locations next to the cylindrical sample and combined to a single mixed sample for each depth. Particle density was measured using pycnometers; particle-size distribution was determined by wet-sieving and sedimentation with Köhn–Pipette method after organic carbon destruction with H₂O₂ and chemical dispersion using Na₄P₂O₇ (Hartge and Horn, 1992). Organic carbon content was measured by gas-analysis (Carbomat 12 D, Wösthoff GmbH, Bochum, Germany).

The soil cylinders were scanned with a medical CT (Siemens Somatom Plus CT scanner, 120 kV, 220 mA) and analyzed as described by Rogasik et al. (2003). Resolution was 1 mm in vertical and 0.25 mm

in lateral direction. The outermost 1 cm was removed to exclude irregular edges and disturbances, thus columns of 8 cm in diameter were analyzed. The CT-images at all depths were examined with respect to rare structures, changes of hue (representing X-ray attenuation) and pattern (e.g., alternating changes) of hue with depth. The visual assessment motivated the following data processing. Data were processed with the software package "Slicer Dicer" (from PIXOTEC, LLC, <http://www.slicerdicer.com>) by assuming a threshold value of -43 HU found by visual inspection to recognize not only air-filled, but also partly-filled biopores and roots and a threshold value of -875 HU for visualization of mostly air-filled regions (Rogasik et al., 2003). Large pores that might not be representatively sampled by the cylinders were retrieved by the two HU-threshold values (i.e., -43 and -875 HU) and a horizontal cross-sectional area of 4 mm² that was defined by visual inspection of the pores. Large pores were removed during analysis of soil structure. Subunits with HU-values below the threshold values were counted for every 2D-image and the areas were calculated from the number of pixels that were connected along a side or a corner. The bulk densities and standard deviations of HU-values were calculated with and without large pores.

The standard deviation (std) of HU-values of the voxels (i.e., a volumetric unit, which is the analogue to a pixel for 2D data) at the given resolution (i.e., 0.25 $\times$ 0.25 $\times$ 1 mm³) was calculated for the horizontal slices to assess their "internal" heterogeneity. The std-values can be markedly dominated by the occurrence of soil structural elements such as macropores or aggregates within segregation structures (Rogasik et al., 2005); high standard deviations of HU-values indicate loosely packed soil horizons or soil horizons with segregation structures, which are characterized by matrix, aggregate

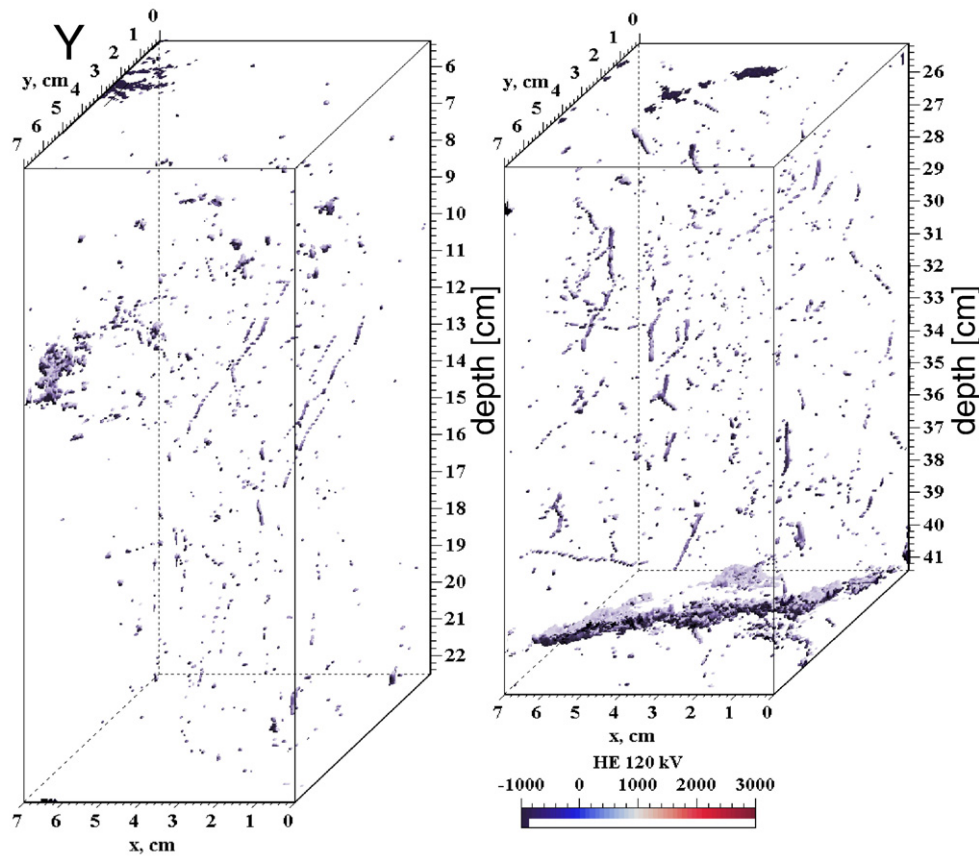


Fig. 5. The 3D distribution of the larger pore system in sample Y, highlighting 'air-filled' regions (HU-values < -875) for 5–22 cm (left) and 25–41 cm depth intervals (right) as in Fig. 3 but with a different HU-threshold value (HE 120 kV).

and macropore composition, and low std-values analogously indicate absence of these structures. Although the std of HU-values allows a characterization of soil structural elements with respect to occurrence and relative size distribution (Rogasik et al., 2003), these values include no further spatial information. Therefore, additional parameters were calculated by a semivariance analysis of images of the 6, 10.8, 15 and 38 cm depths for Y and the 8, 17, 23 and 37 cm depths for O using the proprietary geostatistics software GS+ (Gammadesign, 2007). The active lag distance was set to 66.7% of the total investigated distance. The lag classes were incremented by 1 pixel-length. Within a lag distance of 2 cm, the lag distance with the maximum variance was identified. Then, curves were fitted to the values between 0.08 cm and the lag distance with the maximum variance by minimizing the squared Pearson's coefficient of correlation, r .

The X-ray attenuation of soil depends on the volumetric fraction of the single phases of solid, water, and air with their specific attenuations. The mean values of dry soil bulk density, $\rho_b(z)$ [g cm^{-3}], as a function of soil depth, z [cm], were calculated using measured values of the gravimetric water content, $w(z)$ [g g^{-1}], the mean X-ray attenuation, $\text{HU}(z)$ (air: $\text{HU} = -1000$, water: $\text{HU} = 0$) and the particle densities, $\rho_s(z)$ [g cm^{-3}] at depth z (Rogasik et al., 2003) as:

$$\rho_b(z) = \frac{\rho_s(z)\rho_w[\text{HU}(z) + 1000]}{\rho_w[\text{HU}_s + 1000] + 1000w(z)\rho_s(z)} \quad (1)$$

with

$$\text{HU}_s = \frac{\text{HU}_{\text{core}} + 1000\text{AV}_{\text{core}}}{\text{SV}_{\text{core}}} \quad (2)$$

where ρ_w is density of water [g cm^{-3}], HU_s is the X-ray attenuation of the solid phase, AV and SV are volumetric fractions of air and solid phases, respectively, and subscript *core* denotes the mean sample

value. The vertical distributions of mean dry bulk density and std of HU-values were used to indicate structural changes with depths. The std-values, with exclusion of large pores applying a threshold of -43 HU, were also averaged for each of the four vertical zones that were identified by visual inspection of the bulk density.

For comparison with CT-derived bulk density values, three intact soil cores of 300 cm^3 (6 cm height and 8 cm diameter) were sampled in each of the two paddy fields at the 16–22 cm depth level that should represent the compacted horizon for samples Y and O. After drying for 24 h at 105°C , the dry soil bulk density was determined gravimetrically.

3. Results

The texture is changing with depth (Table 1) as follows: For samples of field Y, the sand fraction is decreasing (from 27.5% to 20.2%), the silt fraction is increasing (37.8% up to 41.2%), and the clay fraction is increasing (34.8% up to 38.6%) with depth (5 cm to 42 cm). The sand fraction is higher and the clay fraction lower for samples from field O as compared to Y. For O, in contrast to Y, the sand fraction first increases with depth (from 34.8% to 41.4%) between 0 and 18 cm and then decreases (27%) down to 35 cm depth, while clay fraction decreases from 0 to 18 cm depth (from 21.4% to 18.4%) and then increases (32.1%) down to 35 cm depth.

The carbon contents (Fig. 2a) decrease with depth from 1.5% (top) to 0.2% (bottom) for sample Y and from 2.1% (top) to 0.3% (bottom) for sample O. In these soils, the total carbon content, C_t , represents the organic carbon, C_{org} , since in all samples, carbonate contents were below analytical detection limit ($<0.01\%$). The values of the gravimetric water content, w , (Fig. 2b) range from 0.22 to 0.34 g g^{-1} for Y and from 0.17 to 0.53 g g^{-1} for O; w is remarkably high in the topsoil of sample O. The mean values of particle densities, ρ_s , (Fig. 2c) show

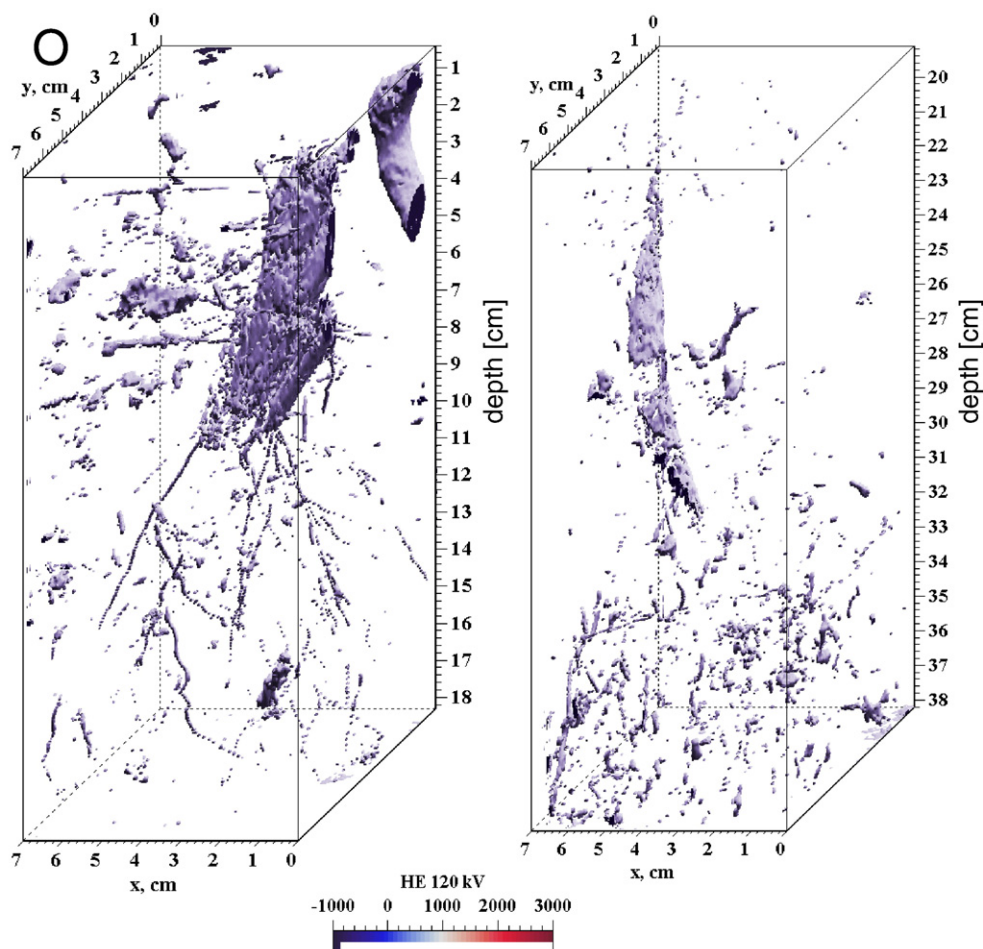


Fig. 6. The 3D distribution of the larger pore system for sample O, highlighting 'air-filled' regions (HU-values < -875) for 0–18 cm (left) and 19–38 cm depth intervals (right) as in Fig. 4 but with a different HU-threshold value (HE 120 kV).

significant (i.e., larger than the std for three repetitions) differences for the samples of the two fields down to 15 cm and beneath 30 cm, with higher ρ_s -values for samples from the younger (Y) as compared to the older paddy field (O). The ρ_s -values for Y increase in the upper 13 cm soil depth from 2.48 g cm^{-3} to 2.57 g cm^{-3} , then decrease with fluctuating values to 2.52 g cm^{-3} at 23 cm and further clearly increase again with depth to a value of 2.61 g cm^{-3} at 39 cm. Values of samples from field O mostly increase with depth from 2.44 g cm^{-3} at the surface to 2.59 g cm^{-3} at about 26 cm depth, but decrease with greater depth (Fig. 2c).

Figs. 3 and 4 show 3D-plots of regions with relatively low X-ray attenuation ('lower density' subunits, HU-values < -43), which include a relatively small portion of regions of lowest attenuation ('air-filled' subunits, HU-values < -875 ; Figs. 5 and 6). Most of the structures identified as 'lower density' subunits that have a width of about 0.5–1 mm are probably formed by partly-filled root channels. In horizontal slices (Fig. 7) these structures are visible as small circle-like black subunits. The central large 'air-filled' and 'low density' structure that is visible in the upper 10 cm of sample O (Figs. 4 and 6) results probably from a rice plant residue. A similarly large structure in the lower part of O, which extends through the compacted pan (20–30 cm depth) and has a cylindrical shape (cf. Fig. 7; 17 and 23 cm depth, right column; Fig. 9, right), might be older since the 'air-filled' structures are less continuous. For sample Y, the number of 'lower density' subunits (Table 2) is 1.12 cm^{-2} (15–31 cm depth) and 1.32 cm^{-2} (31.1–38.2 cm); for sample O, the numbers are between 0.62 cm^{-2} (16–31 cm) and 1.28 cm^{-2} (31.1–38.1 cm). The corresponding numbers of 'air-filled' subunits are 0.11 cm^{-2} and 0.14 cm^{-2} for sample Y and 0.13 cm^{-2} and

0.38 cm^{-2} for sample O. The largest vertical distance between cross-sections without 'air-filled' subunits is 4 mm for Y and 1 mm for O, while 'lower density' subunits (< -43 HU) are continuous for both samples. The mean area of the 'air-filled' subunits (Table 2) is 0.20 mm^2 at the 15.0 to 31.0 cm depth and 0.27 mm^2 (31.1–38.2 cm depth) for sample Y and 0.45 mm^2 (16–31 cm) and 0.48 mm^2 (31.1–38.1 cm) for sample O. Based on the characteristic crack-like shape of the 'low density' subunits, an intensively cracked zone is clearly visible only in sample Y near the interface between the topsoil and the plough pan (about 9.5 to 13.0 cm depth). The crack-like shape is represented by an image at the 10.8 cm depth (Fig. 7, left column; Fig. 8, left).

The soil matrix structure (i.e., the arrangement of soil matrix subunits) is assessed by visual inspection of the distribution of X-ray attenuation in 2D CT-images (Fig. 7) as follows: Structures of more-or-less circular areas of 1 mm to about 20 mm diameter of homogeneous hue can be observed in the gray-scale image for the 6 cm depth (Y). Other 2D images, not presented here, were similarly indicating that these structures continue down to a depth of about 10 cm for Y. The circular structures were not present in images from about 12–22 cm depth (Y) nor in all images for O down to about 31 cm. Fig. 7 shows examples from 15 cm depth for Y and 17 and 23 cm depth for O. In the subsoil (38 and 37 cm depths), relatively comparable structures of about 3–8 mm diameter may be recognized for both O and Y soils. Figs. 8 and 9 visualize the 3D distributions of X-ray attenuation in samples Y and O in the form of selected vertical and horizontal cross-sections. The 2D vertical cross-sections (at $y=2 \text{ cm}$) demonstrate gradual changes with depth of both, the macropore content and structure

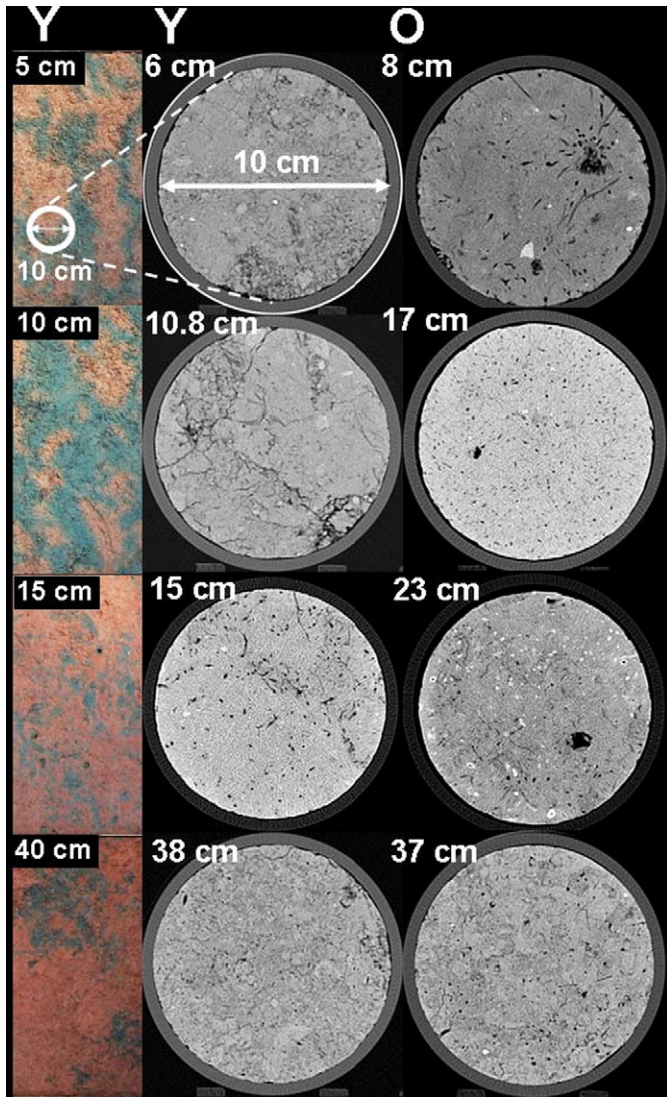


Fig. 7. Horizontal CT-images of the samples from field Y (middle) and O (right) at four selected depths. Horizontal profiles with stained flow paths obtained from Brilliant Blue tracer infiltration experiments at field Y are provided for corresponding depths (left).

and the bulk density changes of the matrix. The 2D horizontal cross-sections demonstrate the lateral continuity at 8 selected depths. For sample Y (Fig. 8), a crack-like pattern is observed at the surface of the plough pan, where the matrix density changes appears relatively gradual. For sample O (Fig. 9), a large opening is dominating the central part above and another macropore below the pan; here the contrasts in matrix densities are more pronounced as compared to Y.

The semivariograms (Fig. 10) are bounded for the investigated depths in the topsoil, the compacted horizons and the subsoil (corresponding images are in Fig. 7). The semivariograms are unbounded within the investigated lag distance for the 10.8 cm depth in the intensively cracked horizon of Y (i.e., the semivariograms further increase for lag distances >2 cm). For distances smaller than 2 cm, a bounded (spherical) model is also fitting to the semivariogram of the 10.8 cm depth. Parameters describing the semivariograms (nugget, sill and range) are given for a lag distance smaller than 2 cm in Table 3.

The semivariograms have the lowest sill in the topsoil at the 6 cm depth for Y and at the 8 cm for O, both with values of $6.4 \times 10^4 \text{ HU}^2$ (Table 3). The highest sill (except for the 10.8 cm depth) is found in both subsoils with $7.6 \times 10^4 \text{ HU}^2$ for Y (38 cm) and $9.1 \times 10^4 \text{ HU}^2$ for O (37 cm). The sill in the pan is relatively similar for Y and O with values of $6.7 \times 10^4 \text{ HU}^2$ for Y (15 cm depth) and $7.0 \times 10^4 \text{ HU}^2$ for O (17 and 23 cm). The nugget accounts for 67% (O, 8 cm) to 94% (Y, 10.8 cm) of the sill. The remaining structure of the variance is highest in the topsoils with about $2.0 \times 10^4 \text{ HU}^2$ occurring within an effective range of 0.7 cm in both images (Table 3). The structures of the variance in the compacted horizons are $0.9 \times 10^4 \text{ HU}^2$ for Y and 0.4 and $1.0 \times 10^4 \text{ HU}^2$ for O occurring within effective ranges of 1.8 cm (Y, 15 cm) and 0.9–1.0 cm (O, 17 and 23 cm). The structures of the variance in the subsoil are $0.9 \times 10^4 \text{ HU}^2$ in Y (38 cm) and $1.0 \times 10^4 \text{ HU}^2$ in O (37 cm), occurring within ranges of 0.6 cm (Y) and 0.5 cm (O).

The vertical distributions of bulk densities calculated from CT data (Eq. (1)) and of the std's of X-ray attenuation for samples Y and O (Fig. 11) are characteristically different. For sample Y, the ρ_b -values increase from 1.21 g cm^{-3} at 7.5 cm depth to maximum values of around 1.63 g cm^{-3} at the top of the compacted horizon (15.3 cm depth). Further downwards, bulk density gradually decreases to 1.35 g cm^{-3} near the 'diffuse' bottom boundary of the compacted horizon (28 cm depth), and eventually approaches a value of $\rho_b = 1.28 \text{ g cm}^{-3}$ in the subsoil. For sample O, the ρ_b -values increase from 1.1 g cm^{-3} at 6 cm depth to a maximum of 1.74 g cm^{-3} (17 cm depth) in the top part of the compacted horizon. Further downwards, bulk density here remains, in contrast to Y, nearly constant throughout the compacted horizon (16–31 cm) at a relatively high level (about $\rho_b = 1.6 \text{ g cm}^{-3}$). At the bottom of sample O (31–35 cm depth), the calculated bulk density rather abruptly decreases to about $\rho_b = 1.4 \text{ g cm}^{-3}$. Depth intervals with contrasting bulk densities can be identified for (i) the topsoil with relatively low values (Y: 53–95 cm depth, O: 6–110 cm depth), (ii) the transition zone with increasing ρ_b -values (Y: 96–149 mm; O: 111–159 mm), (iii) the compacted pan (Y: 150–310 cm; O: 160–311 cm) and for (iv) the transition zone towards the subsoil below the bottom of the compacted zone (Y: 311–381 cm; O: 312–381 cm). For sample Y, the averaged std of HU-values is 281 (with scatter of 67 HU between minimum and maximum values) at the 53 to 95 cm depth interval, 285 HU (scatter: 69 HU) at 96–149 cm, 249 HU (87 HU) at 150–310 cm and 268 HU (43 HU) at 311–382 cm (Fig. 11). For samples of O, the mean HU-values are 228 (scatter: 110 HU) at 6–11 cm, 250 (80 HU) at

Table 2

Numbers and areas of subunits with values below -43 HU representing 'low density' pore regions and below -875 HU representing 'air-filled' pore regions

Sample	Depth interval cm	< -43 HU					< -875 HU				
		Number cm^{-2}			Area mm^2		Number cm^{-2}			Area mm^2	
		mean	min	max	mean	std	mean	min	max	mean	std
Y	^a 15.0–31.0	1.12	0.54	2.13	0.32	0.31	0.11	0.02	0.48	0.20	0.14
	^b 31.1–38.2	1.32	0.86	1.97	0.35	0.39	0.14	0.04	0.34	0.27	0.21
O	^a 16.0–31.0	0.62	1.02	1.55	0.41	0.46	0.13	0.02	0.42	0.45	0.50
	^b 31.1–38.1	1.28	0.10	2.21	0.56	0.65	0.38	0.02	0.78	0.48	0.51

Mean values, minimum (min) and maximum (max) values and standard deviations (std) are calculated from each 2D-image within the depth intervals. Pores $>4 \text{ mm}^2$ were excluded from the analysis.

^aCompacted horizon.

^bSubsoil.

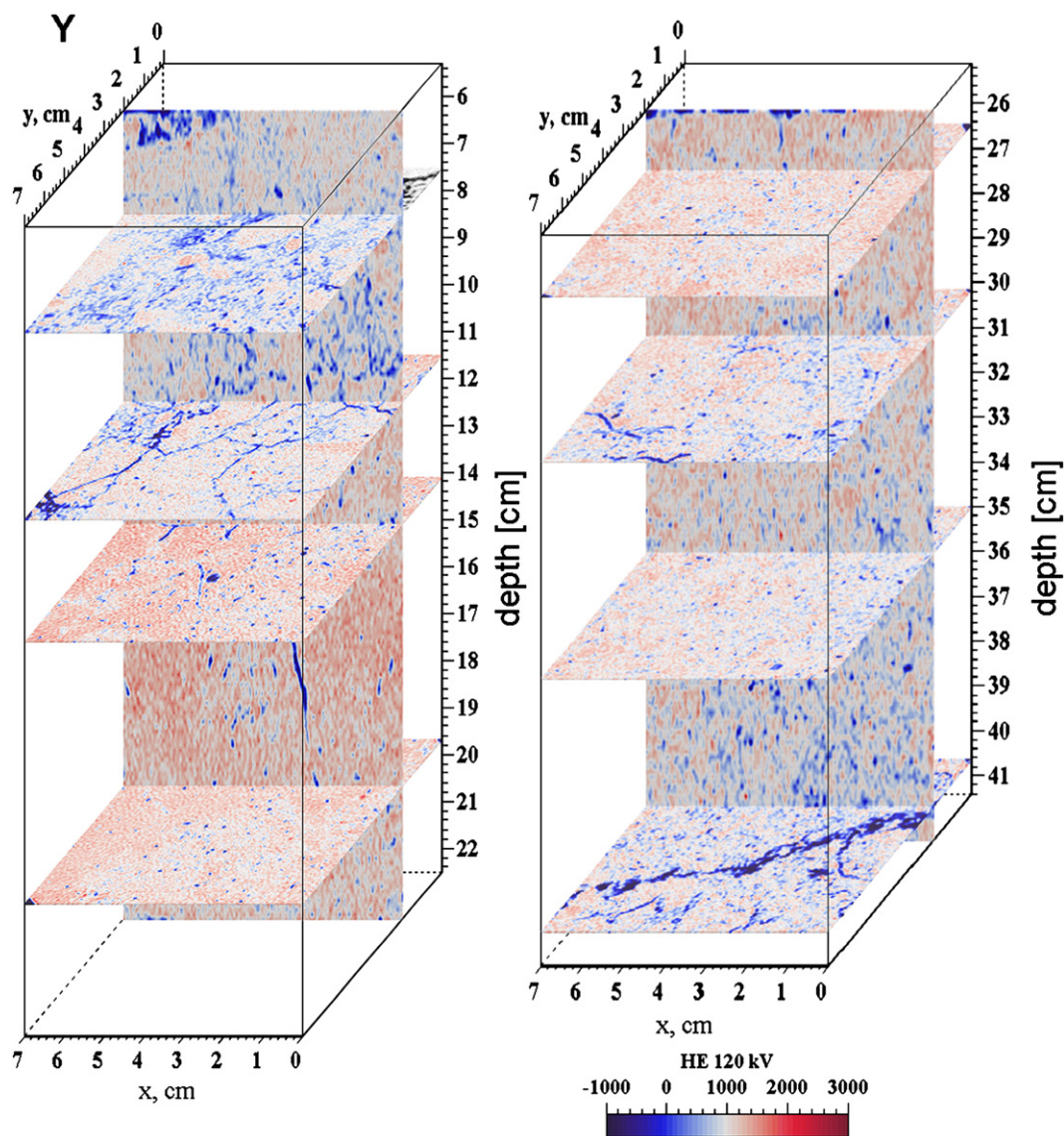


Fig. 8. Image of vertical and horizontal cross-sectional views of 3D X-ray attenuation data for sample Y and depths 5–22 cm (left) and 25–41 cm (right). The scale (bottom) ranges from dark blue for negative HU- (HE 120 kV) values (low density) to dark-red for positive HU-values (high density).

111–159 cm, 261 (113 HU) at 160–311 cm, and 294 (90 HU) at 311–381 cm depth interval (Fig. 11).

The gravimetrically determined soil bulk density for the compacted horizon (16–22 cm depth) was smaller for core samples obtained from field Y (mean: $\rho_b = 1.47 \text{ g cm}^{-3}$, std: 0.01 g cm^{-3}) as compared to core samples from field O (mean: $\rho_b = 1.6 \text{ g cm}^{-3}$, std: 0.01 g cm^{-3}). The relation in bulk densities between Y and O of the core samples compares well with that of the HU-derived values obtained from CT data that are averaged between 16 and 22 cm depth. For sample Y, the mean ρ_b -value of 1.55 g cm^{-3} is 0.08 g cm^{-3} higher than the gravimetrically measured bulk density but considerably smaller as compared to a ρ_b -value of 1.65 g cm^{-3} for sample O, which is about 0.05 g cm^{-3} above than mean obtained from the core samples.

4. Discussion

By means of X-ray computed tomography it was possible to non-destructively characterize the soil structural features of paddy soils that are described below. Despite limitations that rise from few samples, the method can be regarded useful since it allows obtaining quantitative numbers for modelling flow and transport in paddy soils.

In contrast to thin sections or polished blocks, the non-destructive CT-method is more time and cost effective, requires less sample preparation, and yields results soon.

4.1. Root channels

The larger plant residue at 8 cm depth with radial smaller pores (e.g., sample O; Fig. 4) suggest that smaller black mottles (Fig. 7) are root channels. The X-ray attenuation does not allow calculating bulk density at voxel scale, because the HU-value characterizes the wet bulk density; it includes the combined effects of the different constituents (i.e., solid, air and water contents), whereas effects of particle density differences are comparably small. The exact phase distribution at voxel scale is unknown. Thus, it is not feasible to distinguish whether regions of low attenuation ($-875 < \text{HU} < -43$) with dominantly vertical orientation derive from fresh or decayed roots or from loosely- or partly-filled macropores (Figs. 3 and 4). However, the fraction of air-filled macropores can be estimated based on the HU-threshold of -875 (Rogasik et al., 2003) and is given in Figs. 5 and 6. Not all pores of a size that is represented by a voxel might be air-filled; some pores might be filled with organic residues others with water, especially in

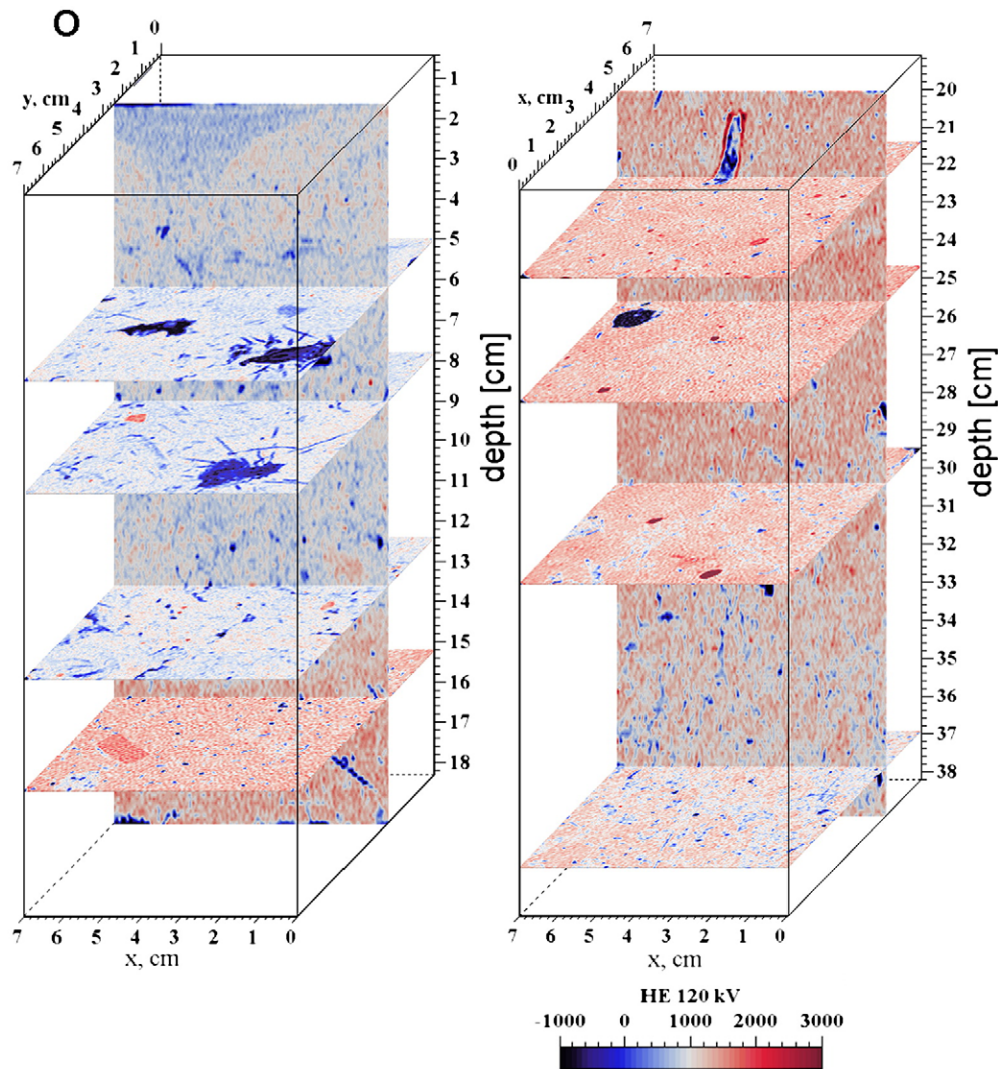


Fig. 9. Image of cross-sectional views of 3D X-ray attenuation data as in Fig. 8 but for sample O and depths 0–18 cm (left) and the 19–38 cm (right). The scale (bottom) ranges from dark blue for negative HU- (HE 120 kV) values (low density) to dark-red for positive HU-values (high density).

the organic-rich topsoil of O due to the relatively high water content of up to 0.53 g g^{-1} (Fig. 2b). Since all samples were gravity-drained, the over-capillary-sized pores can mostly be assumed air-filled. Nevertheless, the number of 'air-filled' pore regions could be somewhat higher than highlighted in Figs. 5 and 6. Any effect of water in larger pores of the voxel scale is expected to be negligible in and below the plough pan where water contents are below 0.32 (Y) and 0.26 (O) g g^{-1} . Air-filled structures ($<4 \text{ mm}^2$) are also located in the vicinity of the large pores in O ($>4 \text{ mm}^2$) indicating that a part of 'air-filled' structures might still be associated with larger pores and that 'air-filled' structures might possibly be overestimated as compared to a sample without larger pores from field O.

Mori et al. (1999) found rice root channels also in his soil samples from Japanese paddy fields and concluded from outflow experiments that these pores induced two-domain bypass flow. The root channels that he presented in thin sections had diameters of about 0.2 mm . In our samples, we find that roots are able to penetrate the compacted plough layer and the adjacent compacted Bw2 horizon, to enter the subsoil and to create partly air-filled vertical pore channels. In the pan, root channels show an even more distinct cylindrical geometry as compared to the channels in the subsoil. The results suggest that these pores strongly affect soil hydraulic properties and water and solute movement in the

paddy soil. Older root channels are likely to facilitate the plant roots to penetrate the soil profile and help improving water and nutrient uptake from the subsoil (Gregory, 2006). Root channels may serve as preferred pathways for air and water flow and solute transport.

4.2. Sampling

It is assumed that the samples represent the structures that are relevant for the investigated scale anywhere in the field. Larger structures, not represented by the samples, such as rare (compared to rice roots) large pores or cracks in the plough pan (as described in Sander and Gerke, 2007) need quantification on the larger scale. However, also smaller-scale structures that are represented by samples of a few cm might vary on plot or field scale. As an example, the distribution of stained pores within the plough pan of a 6.2 m^2 plot was investigated by means of a dye tracer experiment. Two centimeters of Brilliant Blue dye tracer solution was applied directly on the pan and a horizontal profile was created by removing the first $1\text{--}2 \text{ cm}$ of completely stained soil. There was obviously no relation between the small stained pores and the locations of rice stubbles (cf. Fig. 12), indicating that the structure represented by the stained pores is largely independent of the plot scale plant structures.

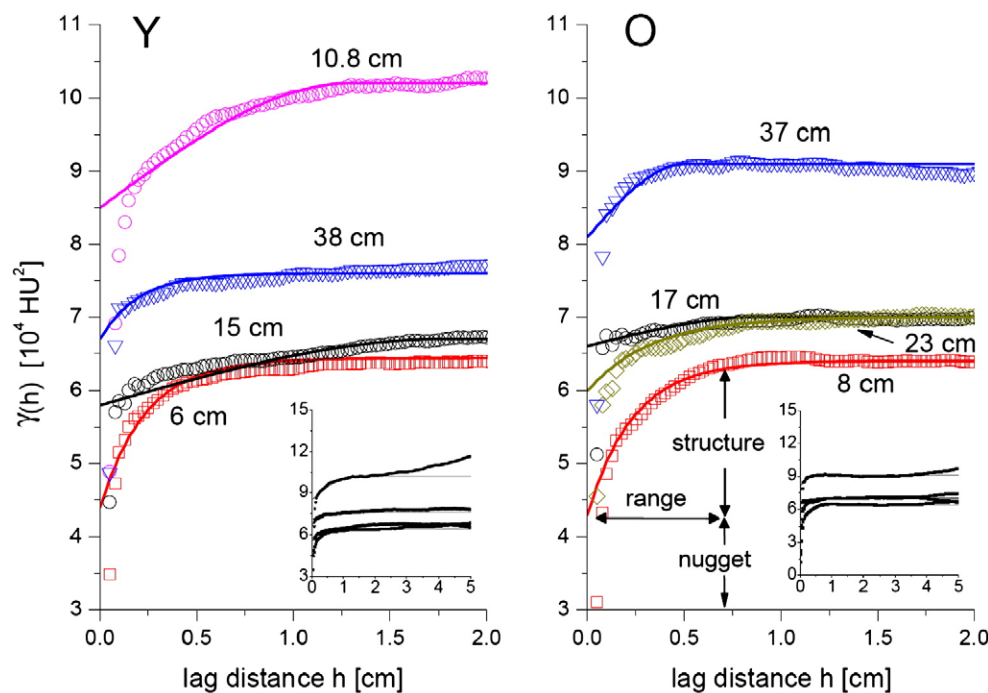


Fig. 10. The semivariograms for distributions of HU-values for depths corresponding to images in Fig. 7. The inset graphs show the semivariograms for distance lags up to 5 cm. Notice, the semivariogram for the 10.8 cm depth is unbounded for lag distances larger than 2 cm. The semivariograms of the 15, 17, and 23 cm depths have a smaller slope (structural semivariance per range) as compared to those of the 6, 8, 37, and 38 cm depths. Corresponding parameters are given in Table 3.

4.3. Inter-aggregate and intra-aggregate spaces

The structures between secondary pores of mostly round areas of 1 to about 20 mm diameter of homogeneous hue in the CT-images (Fig. 7) are interpreted as crumbs formed by aggregation. These aggregates are described for the cultivated horizon of Y and result in a high internal cross-sectional heterogeneity (high std of HU-values), but are not visible for O. The lack of these structures in the topsoil of O results in a smaller heterogeneity in the topsoil as compared to Y, provided that large pores are excluded from the analysis. Probably, aggregate formation took place in field Y, which was cultivated with peanut and not flooded for one cropping period, whereas most aggregates have been destroyed in field O, which has regularly been puddled prior to sampling. The fitted semivariance models do not describe the differences for the two images of the topsoils from fields Y and O. Structures on a small lag distance (e.g., inter-aggregate spaces or root channels) are averaged out on the same range of 0.7 cm. The nugget is not well-suited to describe the differences, because the

model fit is more depending on the semivariogram data close to the range. The fitted nugget values combine all variances of small lag distances. Semivariances for lag distances smaller than the range (Fig. 10) may be described by an additional fitting curve close to the origin of the semivariogram. Alternatively, we provide the std of HU-values for all depths to represent the variance at the pixel scale of 0.25 mm. The mean difference of pairs of classes can also be calculated by multiplying the semivariances by 2 and taking the square root (Nielsen and Wendroth, 2003). The procedure yields average differences in HU-values for a lag distance of 1 pixel of 264 HU for Y (6 cm) and 152 HU for O (8 cm), which indicates that structural differences exist.

The ‘internal’ heterogeneities (i.e., the std of HU-values) in the compacted horizons are higher in the subsoil than in the pan for both samples, corresponding to higher semivariances in the subsoil than in the pan. The semivariances are mostly ascribed to the larger nugget effect, however, also to the larger or equal structural variance in the subsoil that occurs in a significantly smaller range in the subsoil than in the pan and results in a larger slope of the semivariance for the subsoil than for the pan (cf., Fig. 10).

On the one hand, a crumb-like, aggregated structure as found for both subsoil samples (Fig. 7) corresponds with field observations. On the other hand, the blocky and prismatic structure of the compacted horizons, which was recognized during field observations by breaking of samples, is not reflected in CT data. This discrepancy indicates that the inter-aggregate pore space between macro-aggregates is probably too small to become visible in the images considering the CT resolution.

4.4. Effects on hydraulic processes

The soil structure correlates with dye tracer pattern of Sander and Gerke (2007) for field Y (see Fig. 7). The flow in the compacted horizon with relatively homogeneous soil matrix structure was confined to the direct vicinity of large macropores while such bypass flow was smaller in the aggregated top- and subsoil horizons where the lateral water

Table 3

Parameters describing the semivariograms (Fig. 10) within a lag distance of 2 cm for horizontal images at given depths (corresponding images are in Fig. 7) for the fitted models with Pearson's squared coefficient of correlation, r

Sample	Depth interval cm	Nugget	Sill	Structure	Effective range cm	r^2	Model
			10^4 HU ²				
Y	6	4.4	6.4	2.0	0.7	0.99	Exponential
	10.8	8.5	10.2	1.7	1.3	0.97	Spherical
	15	5.8	6.7	0.9	1.8	0.96	Spherical
	38	6.7	7.6	0.9	0.6	0.86	Exponential
O	8	4.3	6.4	2.1	0.7	0.99	Exponential
	17	6.6	7.0	0.4	0.9	0.91	Spherical
	23	6.0	7.0	1.0	1.0	0.96	Exponential
	37	8.1	9.1	1.0	0.5	0.96	Exponential

exchange from macropores into the surrounding matrix was comparably large. In this study, most of the soil volume of the compacted horizon was bypassed, which resulted in fast and deep movement of dye tracer to the subsoil and indicated possible risk for groundwater quality by leaching of fertilizers or pesticides.

An intensively cracked zone between the cultivated (puddled) soil horizon and the plough pan is demonstrated in the CT data for sample Y (Fig. 7), and was also observed in the soil profiles of field Y when it was not flooded. These structures might cause lateral preferential flow on the plough pan as measured by Janssen and Lennartz (2007). The cracked zone is not recognized for sample O, which was sampled only 3 days after draining the saturated topsoil. The large crack that is visible at the bottom (41 cm depth) of sample Y (Fig. 8) resulted from disturbance during sampling.

4.5. Bulk density distribution

The gravimetric water content depends on conditions during sampling and storing in the aluminium box; for instance, condensation of water may have affected water contents at the bottom and top parts of the monoliths. Here, the water content variability with depth is included in Eqs. (1) and (2) for calculation of the bulk density.

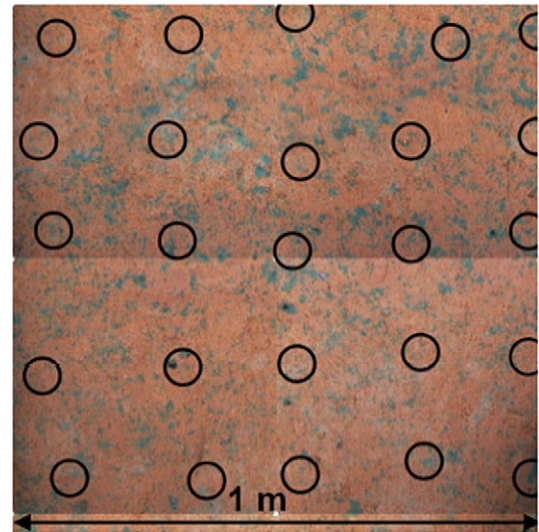


Fig. 12. Horizontal paddy-soil profile excavated at 1–2 cm below the surface of the plough pan. The blue dye-stained areas indicate flow pathways and the black open circles mark the locations of rice stubbles. Note that excavation followed the uneven topography of the plough pan surface with its alternating depths.

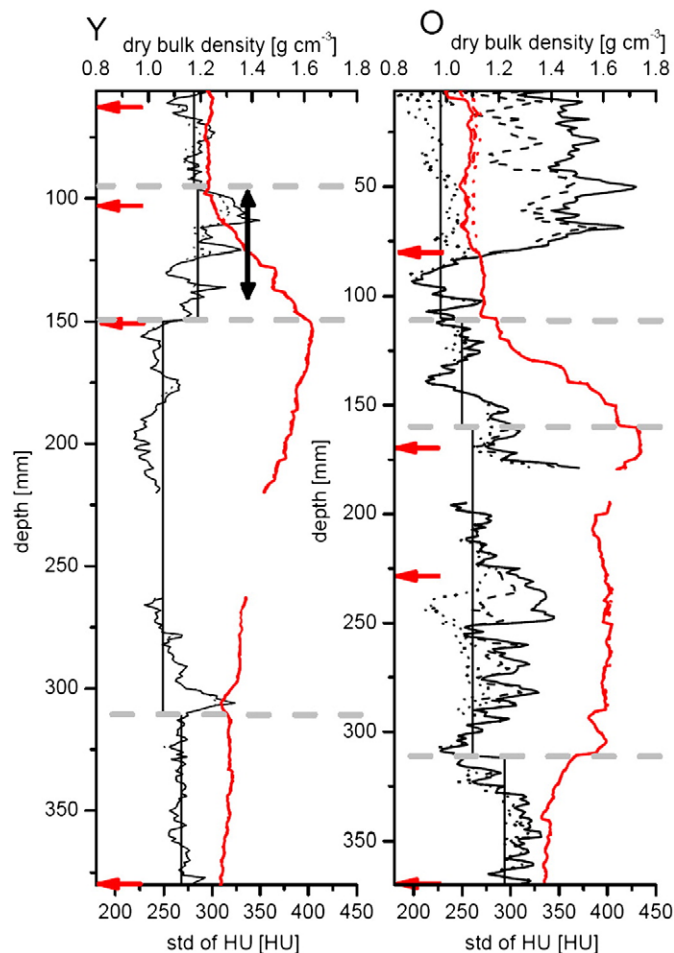


Fig. 11. Calculated dry bulk density (red lines) and standard deviation (std) of Hounsfield Units (HU) (black lines) as a function of soil depth for samples Y (left) and O (right); std-values after exclusion of pore regions $>4 \text{ mm}^2$ with HU-values <-875 (black dashed lines) and HU-values <-43 (black dotted lines). Gray dashed lines indicate the boundaries between topsoil, transition zone towards plough pan, compacted horizon and subsoil. Red arrows mark depths of horizontal images (Fig. 7). The black vertical arrow indicates an intensively cracked zone.

Calculated bulk density values and measured values obtained from core samples from the pan agree well and demonstrate that the core sample averages do not represent the local maximum bulk density of the pan. Calculated bulk densities differ for the samples from the two fields, with the highest value for sample O being 0.11 g cm^{-3} larger than the value of the sample (Y) from the younger paddy field. The comparably less compacted pan of the younger paddy soil will probably be more susceptible to future land use and soil managements despite more than 20 years of compaction by ploughing and puddling with water buffalos and cows. All ρ_b -values for samples from field Y and, beneath 31 cm, also for sample O, are far below the maximal recognized dry bulk density value of $\rho_b = 1.74 \text{ g cm}^{-3}$ (O). Also for the subsoil, ρ_b is significantly higher in the sample from field O (at 37 cm depth) than in that from field Y suggesting that bulk density may have increased also beneath the intensively compacted layer by long-term effects of soil cultivation. Thus, especially for the younger paddy field, the data indicate that the soil might have not yet reached the ρ_b -values that are likely possible by long-term mechanical compaction.

Even if absolute values of the individual samples are ignored, a comparison between the relative differences in vertical distributions of bulk densities may be interesting. Compaction gradually decreases with depth in the pan of the soil from the younger paddy field but remains almost constant in that from the older one down to 31 cm. The latter probably results from long-term impact of tillage but may also be affected by compaction during shrinkage and swelling and by chemical cementation (Sharma and De Datta, 1986).

Despite not directly included in the study, we want to stress that the effects of increasing mechanization concerning structure of paddy soils, should be carefully tested. Heavier machines compared to buffalos may accelerate and increase compaction and lead to a change in pore and matrix structures. The results of this study may contribute to improve the knowledge on water flow, soil/plant interactions, and soil quality in paddy soils and may help in stimulating suggestions for further investigations.

5. Conclusions

The X-ray CT-scanning technique allows the visualization and description of air-filled pores, partly-filled pore regions, and aggregation

of soils from paddy fields. Findings for different depths and samples of two soils are discussed by quantitative queries on the size and number of 'air-filled' and 'low-density' pore structures, internal heterogeneities and semivariations. The X-ray scanning results together with water contents and particle densities at different depths allow calculating the vertical mean bulk density distribution. Descriptions of the soil structure need detailed discussion, despite quantitative queries. The results for soil samples of both fields, O and Y, indicate the existence of vertical root channels in the compacted horizons and of a crumb-like structure in the subsoil. Differences that might be due to previous puddling are the absence of an intensively cracked zone and absence of the aggregated topsoil in sample O as compared to Y. Differences that might be due to different duration of paddy rice cultivation of the fields (O: 100 years; Y: 20 years) are a larger and more uniform bulk density in the compacted horizon that continues with depth in sample O as compared to sample Y. The morphology of vertical root channels, interfacial cracks, and inter- and intra-aggregate pore spaces might strongly affect soil hydraulic properties and may allow preferential water and solute movement in the paddy soils.

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